Template Document

Academic Support Agent

February 19, 2025

1 General Instructions

Please use this template to prepare your manuscript for submission to the Journal for Modeling in Ophthalmology. You will find here many useful instructions on how to organize and format your manuscript. The main body of the manuscript should be contained in a .tex file and the bibliography in a .bib file. The .tex file, the .bib file and the class file **jmoarticle.cls** should be located in the same folder. For the ease of reference, you are strongly encouraged to name the .tex and .bib files as follows:

LASTNAME_FIRSTNAME_JMO.tex
LASTNAME_FIRSTNAME_JMO.bib

Please use the class file jmoarticle.cls with all default options and please do not alter the class file jmoarticle.cls under any circumstances. Should you experience any problem in utlizing the class file, please prontly contact Kugler Publications at:

info@modeling-ophthalmology.com

2 Abstract

Please provide an abstract of 350 words or less. The abstract must be structured within the four subheadings above, namely *Purpose*, *Methods*, *Results* and *Conclusions*. The abstract should be informative and appealing to an interdisciplinary audience.

3 Keywords

Please provide 5 to 8 keywords or phrases in alphabetical order, separated by commas.

4 Text formatting

4.1 Spelling

Please set spelling to English-US.

4.2 Abbreviations

Please use only abbreviations that are common and necessary to the exposition of the text. Abbreviations should be widely understood by the JMO multi-disciplinary audience that includes ophthalmologists, physiologists, mathematicians, engineers and physicists. Introduce each abbreviation in parentheses after the first use of the full term. Avoid abbreviations that have meaning only within the context of the specific manuscript.

4.3 Hyphenation

Please type your text with hyphenation off.

4.4 Hidden text

Do not use hidden text such as endnotes, footnotes, index entries, figure boxes, links, hyperlinks, bookmarks, references or table boxes. Provide us with plain text and tables only.

4.5 Units

Please use SI units of measure (http://physics.nist.gov/cuu/Units/units.html). Please write 20 mm or 4 mmHg with spaces, and not 20mm or 4mmHg.

4.6 Latex packages

Additional Latex packages, if needed, should be added in the preamble of the .tex file. Please do not alter the class file .cls and the formatting style under any circumstances. Should additional required packages lead to compiling errors, please contact Kugler Publications at:

info@modeling-ophthalmology.com

4.7 Special text

Please write latin terms (medical) in italic. For example *ophtalmos*, *occulus*, ...

5 Organization of the Manuscript

5.1 Suggested sections

You are strongly encouraged to organize the manuscript into five main sections entitled *Introduction*, *Methods*, *Results*, *Discussions*, *Conclusions and future perspectives*.

5.1.1 Introduction

Please provide an informative overview of the background motivating your work. The introduction of your manuscript is particularly important as it helps creating a forum among scientists of different expertise. Please be sure to cite the relevant literature and to clearly identify open questions and/or controversial issues in the field.

5.1.2 Methods

Please provide a detailed description of the methods utilized in your work. JMO articles may be very diverse and involve experimental set-ups, clinical studies, statistical analysis, mathematical modeling and/or numerical simulations. Regardless of its nature, every JMO article is expected to provide all the details that are necessary in order to reproduce the work described in the article. the use of Appendices or Supplemental Material is also allowed to accommodate lengthy but useful method descriptions.

5.1.3 Results

Please provide a clear description of the results. Follow the same order as that of figures and tables referred to in the text. Please leave the result discussion for the next section.

5.1.4 Discussion

Please provide a thoughtful discussion of the results. Recall the main questions you aimed at addressing in your work. Critically compare your findings with those obtained by others and provide possible explanations for observed differences and/or similarities. Clearly identify the limitations in your work and argue how and to what extent they might influence the results.

5.1.5 Conclusions and future perspectives

Please provide a brief summary (up to two sentences) of your work and identify directions of future development of your work that could be addressed using interdisciplinary approaches.

5.2 Tables

Authors must use a specific table format to respect JMO design. The JMO article class provide a specific color for table headers named jmotablecolor. See the following example

```
\begin{table}[h]
  \centering
  \begin{tabularx}{\textwidth}{|p{.5\linewidth}|X|}
    \hline
    & \cellcolor{jmotablecolor} description \\
    \hline
    \cellcolor{jmotablecolor} Field 1 & 42 \\
    \cellcolor{jmotablecolor} Field 2 & 42 \\
    \hline
  \end{tabularx}
  \caption{Format table with JMO colors result.}
  \label{tab:tab1}
  \begin{flushleft}
    Add subnote for this table
  \end{flushleft}
\end{table}
```

which produces the result seen in table tab:tab1. An optional macro \ tabref is provided to cite table.

	jmotablecolor description
jmotablecolor Field 1*	42
jmotablecolor Field 2	42

Table 1: Format table with JMO colors result.

*Add subnote for this table

5.3 Figures

The resolution of all image files should be 300 DPI or higher. Not only will lower resolutions result in poor print quality, but also new, higher-resolution screens such as retina displays, result in a poor-quality figure. The font size of all symbols and characters included in the figures must be large enough to be readable. There is no additional fee for the use of color figures. Captions should be informative and deliver the main message represented in the figure. All symbols or letters that appear in the figures should be defined in the caption. Please seek permission for the use of figures from other publications and provide the source. Figures can be included using the command `includegraphics` as in the example in fig:Example$_{f}ig$:
A optional macro \ figref is provided to cite figures. A second way to create your own figure is to using `tikz` as in the example in fig:kirchoff$_{v}oltage$

6 Code listings

You can insert a sample of source codes using the listings package. Do not alter the default font or color formatting under any circumstances. [caption=Example of source code,style=latex]

Listing 1: C++ source code

```
int main()
{
    std::cout << "Example␣of␣code\n";
    return 0;
}
```

The result of the previous code can be seen in codeexample.

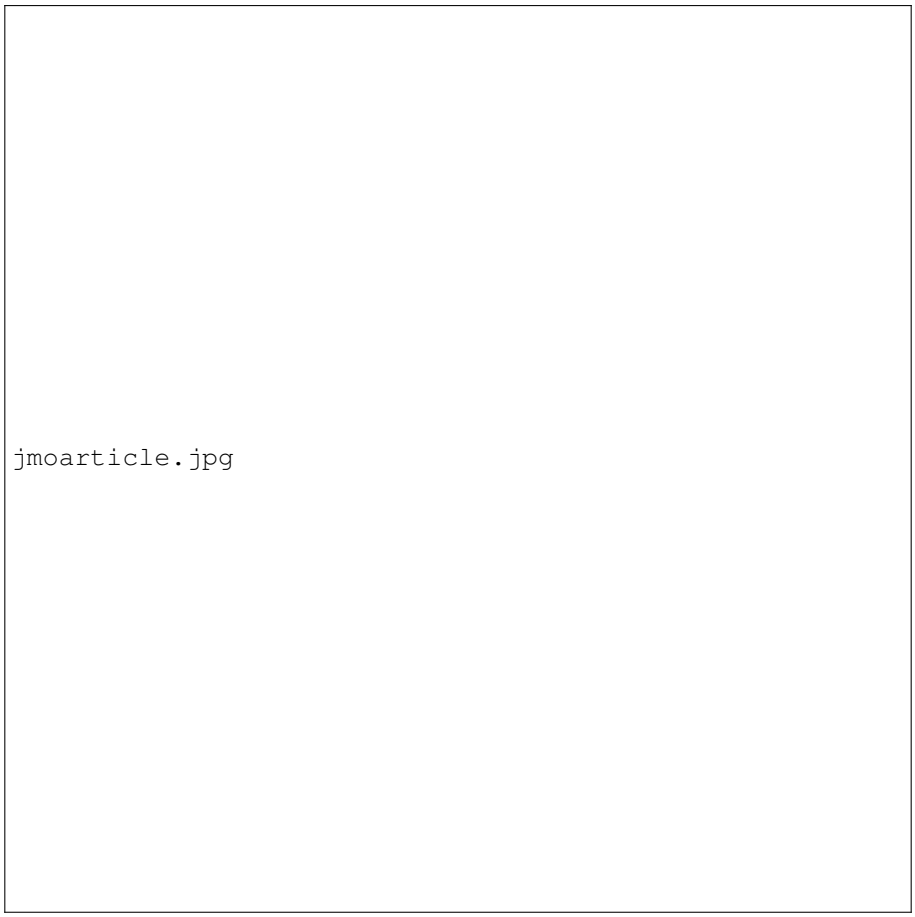


Figure 1: Example of an image included using the command `includegraphics`. The image has been reproduced from the article by Carichino et al [**?**] upon permission of Kugler Publications.

[fill] (0,0) circle [radius=0.05]; [above left] at (0,0) A; (-2.2,0) – (0,0); [dashed] (-3,0) – (-2.2,0); [->] (-1.2,0.2) – (-0.5,0.2); [above] at (-0.8,0.2) i_3; (-1.5,-1.5) – (0,0); [dashed] (-2,-2) – (-1.5,-1.5); [<-] (-1.3,-1) – (-0.7,-0.4); [above] at (-1.1,-0.7) i_2; (0,-1.5) – (0,0); [dashed] (0,-2) – (0,-1.5); [->] (0.2,-1) – (0.2,-0.4); [right] at (0.3,-0.6) i_1; (0,0) – (1.2,0); [scale=0.20] [shift=(6,0),rotate=0] (0,0) –(0.5,0.866025404) – (1.5,-0.866025404) – (2.5,0.866025404) – (3.5,-0.866025404) – (4,0); (2,0) – (3,0); [dashed] (3,0) – (4,0); [->] (0.3,0.2) – (1,0.2); [above] at (0.9,0.2) i_5; [fill] (3,0) circle [radius=0.05]; [above left] at (3,0) D; (1.2,-0.3) circle [radius=0.15]; at (1.2,-0.3) $+$; [<-, very thick, blue] (1.2,-0.5) arc [radius=0.5, start angle=200, end angle= 340]; [below,blue] at (1.7,-0.2) v_5; (2.1,-0.3) circle [radius=0.15]; at (2.1,-0.3) $-$; (0,0) – (0,1); (0,1.3) – (0,2.3); (-0.4,1) – (0.4,1); (-0.4,1.3) – (0.4,1.3); [->] (0.2,0.2) – (0.2,0.8); [right] at (0.2,0.6) i_4; [fill] (0,2.3) circle [radius=0.05]; [above left] at (0,2.3) B; (-0.5,1.5) circle [radius=0.15]; at (-0.5,1.5) $-$; [->, very thick, blue] (-0.7,1.7) arc [radius=0.5, start angle=110, end angle= 250]; [blue] at (-0.7,1.1) v_4; (-0.5,0.8) circle [radius=0.15]; at (-0.5,0.8) $+$; (0,2.3) – (1.2,2.3); (1.2,2.7) circle [radius=0.15]; at (1.2,2.7) $+$; [->, very thick, blue] (2.1,2.9) arc [radius=0.5, start angle=20, end angle= 160]; [above,blue] at (1.6,2.7) v_6; [scale=0.20] [shift=(6,11.5),rotate=0] (0,0) –(0.5,0.866025404) – (1.5,-0.866025404) – (2.5,0.866025404) – (3.5,-0.866025404) – (4,0); (2,2.3) – (3,2.3); (2,2.7) circle [radius=0.15]; at (2,2.7) $-$; [dashed] (3,2.3) – (4,2.3); [->] (0.3,2.1) – (1,2.1); [below] at (0.7,2.1) i_6; [fill] (3,2.3) circle [radius=0.05]; [above left] at (3,2.3) C; (3,0) – (3,1); (3,1.3) – (3,2.3); (2.6,1) – (3.4,1); (2.6,1.3) – (3.4,1.3); [->] (3.2,0.2) – (3.2,0.8); [right] at (3.2,0.6) i_7; (3.7,1.5) circle [radius=0.15]; at (3.7,1.5) $-$; [<-, very thick, blue] (3.9,0.7) arc [radius=0.5, start angle=-70, end angle= 70]; [blue] at (3.9,1.1) v_7; (3.7,0.8) circle [radius=0.15]; at (3.7,0.8) $+$; [->, very thick] (1.3,0.8) arc [radius=0.5, start angle=240, end angle= 30];

Figure 2: Schematic representation of a lumped electric circuit. The circuit nodes are denoted by the capital letters A, B, C and D. Voltages across capacitors and resistors are denoted by v_i, with $i = 4, \ldots, 7$. Currents are denoted by i_k, with $k = 1, \ldots, 7$.

Listing 2: C++ source code

```
int main()
{
    std::cout << "Example of code\n";
    return 0;
}
```

An optional macro \codref is provided to cite lstlisting codes.

7 Bibliography

7.1 How to write the bibliography file ?

The bibliographic items must be entered in a file with extension .bib, such as that used in this template. The file should be located in the same folder as the .cls and .tex files. The bibliography uses the Vancouver style. You are encouraged to pay particular attention to the way in which you fill the .bib file. In particular, please check the following items:

- title : the title should be in lower case letters, except for particular words that do require capital letters, such as the first word or capitalized words required in the title.
- authors: authors listed in the bibliography file should be separated by "and". Lastname and firstname should be separated by a comma in the given order:

  Lastname1, Firstname1 and Lastname2, Firstname2

  First character for the firstname is also allowed:

  Lastname1, F and Lastname2, F

  Note that the firstname is always abbreviated in the document [**?**].
- journal: use the common abbreviated name for the Journal and not its full name. For example, write "Prog Retin Eye Res" instead of "Progress in Retinal and Eye Research".
- url: provide an url/link pointing to the article. The link must point to the official journal where the article is published!

Examples of correct bibliographic items are provided within this template (see "jomarticle.bib" for more details) for:

- a journal article [**?**];
- an article in conference proceeding [**?**];
- an authored book [**?**];
- an edited book [**?**];
- a book within a collection [**?**];
- a book chapter [**?**].

7.2 Special bibliography fields ?

The JMO article class provides for authors two specific bibliography fields for PubMed articles

- pmid: PubMed identifier or PubMed unique identifier
- pmcid: PubMed Central reference number

We strongly encourage authors to fill these fields when identifiers are available!

[pmid, pmcid] Note that pmid and pmcid identifiers are printed out only using the bibliography compiler (not with which is used by default). Note that editors will use for the final version.

[biber] If you are interested in using , you have to enable the backend. Add to the class options

```
\documentclass[biber]{jmoarticle}
```

Remove temporary ".aux" and ".bbl" generated files if they exist before recompiling the latex document. For author(s) who are used to command line, proceed with the following commands

```
pdflatex jmoarticle.tex
biber jmoarticle.bcf
pdflatex jmoarticle.tex
```

[bibtex] For bibtex user, just go with the default document class.

```
pdflatex jmoarticle.tex
bibtex jmoarticle.aux
pdflatex jmoarticle.tex
```

[compilers compatibility] The class is compatible with most common latex compiler (pdflatex, xelatex, lualatex, ...). For any problem please contact the editor

info@modeling-ophthalmology.com

8 Troubleshooting

For any problem please contact the editor

info@modeling-ophthalmology.com

Acknowledgements

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